



Software Requirement Collection Document (SRCD)

Project Title

AI-Powered Multi-Disease Chest X-Ray Triage and Clinical Decision Support System

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Submitted by:

Zinnatun Tasnim (232014024)

Rafsan Janin washi (233014019)

Abdullah Al Tanmim (233014185)

Shihab Ahmed Sakib (231014148)

1. Introduction

Chest X-rays are one of the most commonly used medical imaging techniques for diagnosing lung and thoracic diseases. Although deep learning models have achieved impressive accuracy in detecting diseases from X-ray images, many existing systems fail to provide clear explanations for their predictions and do not assist users in making practical decisions.

This project aims to develop an AI-powered clinical decision support system that can analyze chest X-ray images, detect multiple thoracic diseases simultaneously, and provide visual explanations using Explainable AI (XAI). The system will assist both patients and healthcare professionals by offering disease predictions, urgency assessments, and diagnostic support while ensuring that final medical decisions remain with qualified healthcare providers.

2. Problem Identification

The current healthcare system faces several challenges in chest X-ray diagnosis:

- Manual interpretation of X-rays can be time-consuming.
- Many AI-based solutions operate as "black-box" systems without explaining their predictions.
- Multiple diseases may occur simultaneously, making diagnosis more complex.
- Existing systems often provide only prediction scores without meaningful clinical guidance.
- Patients in remote areas may not have immediate access to radiology experts.

As a result, there is a need for an intelligent, explainable, and user-friendly system that can support disease detection and triage decisions.

3. Project Objectives

The main objectives of the project are:

1. To develop a deep learning model capable of detecting multiple thoracic diseases from chest X-ray images.
2. To provide visual explanations of AI predictions using Grad-CAM heatmaps.
3. To build a mobile application for easy image upload and result visualization.
4. To develop a backend API that performs image processing and disease prediction.
5. To implement a rule-based triage engine that categorizes cases based on urgency.
6. To support clinicians by providing diagnostic suggestions and supporting evidence.
7. To evaluate the system's performance using standard machine learning metrics.

4. Scope of the Project

In Scope

- Uploading chest X-ray images.
- Automated disease prediction using deep learning.
- Multi-label disease classification.
- Generation of Grad-CAM heatmaps.
- Patient dashboard for viewing results.
- Clinician dashboard for detailed analysis.
- Rule-based triage recommendations.
- Secure communication between mobile app and backend server.

Out of Scope

- Direct prescription of medications.
- Replacement of professional medical diagnosis.
- Integration with hospital Electronic Health Record (EHR) systems.
- Real-time monitoring of patients.
- Processing imaging modalities other than chest X-rays (CT, MRI, ultrasound, etc.).

5. Feasibility Study

Technical Feasibility

This project is technically achievable using currently available tools, frameworks, and publicly accessible datasets. The development team has prior experience with programming, machine learning, and mobile application development, which will help in implementing the proposed system.

The main technologies planned for this project are:

- Python for model development and backend services
- FastAPI for API development
- Flutter for mobile application development
- PyTorch for deep learning model training
- OpenCV for image preprocessing

- Docker for deployment and environment management
- NIH ChestX-ray14 or CheXpert dataset for training and evaluation

Since these technologies are widely used and well documented, sufficient learning resources and community support are available throughout the development process.

Economic Feasibility

The project does not require significant financial investment. Most of the software tools, libraries, and datasets needed for development are available free of cost under open-source licenses.

The development and testing activities will be carried out using existing personal computers and university laboratory resources. Therefore, no additional expenditure for specialized hardware or software licenses is expected during the project period.

Operational Feasibility

The proposed system is intended to be simple and practical for both patients and healthcare professionals. Users will be able to upload chest X-ray images through a mobile application and receive prediction results in an easy-to-understand format.

For clinicians, the inclusion of visual explanations such as heatmaps can help them better understand the model's predictions. Since the system is designed as a decision-support tool rather than a replacement for medical experts, it can be integrated into existing diagnostic workflows with minimal difficulty.

Schedule Feasibility

Considering the project scope and the duration of the academic semester, the proposed work can be completed within the available timeframe. The project activities will be divided into several stages, including requirement analysis, dataset preparation, model development, application development, testing, and documentation.

By distributing tasks among team members and following a planned schedule, the project is expected to be completed successfully within the semester.

6. Stakeholders and Users

The following stakeholders are directly involved in or affected by the proposed system

Stakeholder	Role in the Project
Patients	Use the mobile application to upload chest X-ray images and view the generated results and recommendations.
Doctors/Clinicians	Review the AI-generated predictions, heatmaps, and supporting information to assist in clinical decision-making.
Project Team	Responsible for designing, developing, testing, and maintaining the system throughout the project lifecycle.
Faculty Supervisor	Provides guidance, monitors project progress, and evaluates the overall outcome of the project.

7. Requirement Collection

To gather the requirements for this project, our team followed several approaches. Since the project is related to healthcare and medical image analysis, we focused on understanding both the technical requirements and the practical needs of potential users. The requirements were collected through discussions, research, and analysis of existing systems.

Sources of Requirements

Stakeholders

The primary stakeholders identified for this project are:

- Healthcare professionals who regularly work with chest X-ray reports.
- Patients who may use the system to obtain preliminary screening results.
- Faculty supervisors who provide academic guidance and project evaluation.
- Project team members responsible for designing and implementing the system.

Documents and Research Materials

To understand the problem domain and existing solutions, we reviewed:

- Research papers related to chest X-ray disease detection using deep learning.
- Documentation of publicly available medical imaging datasets such as NIH ChestX-ray14 and CheXpert.
- Articles and reports on Explainable Artificial Intelligence (XAI).
- Existing AI-assisted diagnostic applications and clinical decision support systems.

Existing Systems

We also studied several existing systems to understand their strengths and limitations. This helped us identify features that could improve usability and reliability in our proposed solution.

- Clinical decision support systems used in healthcare environments.
- Deep learning-based medical image classification platforms.
- AI-powered diagnostic assistance tools available in research and industry.

Requirement Collection Techniques

Discussion and Consultation

The team discussed the project idea with faculty members and consulted available medical resources to better understand the workflow of chest X-ray diagnosis. These discussions helped us identify the key functionalities required in the system.

Literature Review

A detailed review of journal articles, conference papers, and technical reports was conducted. This helped us understand current research trends, commonly used deep learning models, and the importance of explainable AI in healthcare applications.

Observation and Analysis

We observed how chest X-ray reports are generally interpreted and how diagnostic decisions are made. This helped us understand the need for visual explanations and urgency-based recommendations.

Prototype Planning

Initial interface sketches and workflow diagrams were prepared to visualize how users would interact with the system. Feedback from team members and supervisors was considered while refining the requirements.

8. Functional Requirements

Functional requirements describe the services and features that the system must provide.

Module	Requirement Description
User Management	The system shall allow users to create an account and log in securely. The system shall support different user roles such as patient and clinician. Users shall be able to update their profile information when necessary.
Image Upload and Processing	Users shall be able to upload chest X-ray images through the mobile application. The system shall verify whether the uploaded file is a valid image before processing. The system shall preprocess the image before sending it to the prediction model.
Disease Detection	The system shall analyze uploaded chest X-ray images using a trained deep learning model. The system shall support multi-label classification for detecting multiple thoracic diseases. The system shall generate confidence scores for each predicted disease.
Explainable AI Features	The system shall generate Grad-CAM heatmaps to highlight important regions of the X-ray image. The generated heatmap shall be displayed together with the prediction results. The system shall provide a brief explanation of the prediction output for easier understanding.

Clinical Decision Support	The system shall categorize cases into different urgency levels based on prediction results. The system shall provide supportive recommendations that may assist healthcare professionals during evaluation. The system shall clearly indicate that the prediction is generated by AI and should not replace professional medical judgment.
Reporting and History	Users shall be able to view previous prediction records. Clinicians shall be able to access detailed prediction information and visual explanations. The system shall maintain a history of uploaded images and generated reports.

9. Non-Functional Requirements

Non-functional requirements define the quality attributes and operational characteristics of the system.

Category	Requirement Description
Performance	The system should generate prediction results within a reasonable time, preferably within 10 seconds. The system should be capable of handling multiple user requests simultaneously. Image upload and processing should be completed without noticeable delays under normal conditions.
Security	All communication between the mobile application and server should use secure protocols such as HTTPS. User credentials should be stored securely using encryption or hashing techniques. Medical images and user information should be protected from unauthorized access.

Reliability	The system should provide consistent prediction results for valid inputs. The application should remain operational with minimal downtime. In case of failure, the system should display appropriate error messages and recover gracefully.
Usability	The application interface should be simple and easy to understand. Users should be able to upload images and view results without requiring technical knowledge. Prediction results should be presented in a clear and readable format.
Scalability	The system should support future integration of additional disease categories. The architecture should allow future deployment on cloud platforms if required.

10. Data Requirements

The system will manage several categories of data during operation.

Patient Information

The following information may be stored for user management purposes:

- Patient ID
- Full Name
- Email Address
- User Role
- Registration Date

Chest X-Ray Information

For each uploaded image, the system will maintain:

- Image ID
- Upload Date and Time
- Image File Location
- Processing Status

Prediction Information

The prediction module will generate and store:

- Prediction ID
- Detected Disease Labels
- Confidence Scores
- Triage Category
- Generated Heatmap Reference
- Prediction Timestamp

11. Assumptions and Constraints

Assumptions

The following assumptions were considered during requirement analysis:

- Users will upload chest X-ray images in supported formats.
- Internet connectivity will be available while using the application.
- Publicly available datasets will provide sufficient data for model training and evaluation.
- Users understand that the system provides decision support rather than a final diagnosis.

Constraints

The project also has several limitations and constraints:

- Prediction accuracy may vary depending on the quality of the uploaded X-ray image.
- Training deep learning models requires significant computational resources and time.

- The system is intended for educational and research purposes and should not be considered a replacement for professional medical consultation.
- Access to real clinical data may be limited due to privacy and ethical considerations.

13. Conclusion

The proposed AI-Powered Multi-Disease Chest X-Ray Triage and Clinical Decision Support System aims to bridge the gap between advanced deep learning research and practical healthcare applications. By combining multi-disease detection, explainable AI, and a user-friendly mobile platform, the system will provide valuable support to both patients and healthcare professionals. The project is technically feasible, economically affordable, and achievable within the available academic timeline.